PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Document register and engineering queries

PVC-V1-PM-DOC-001

The register of the engineering documents generated for R-101 by the PVC Suspension Reactor Ver 1 OTS (19 entries), the map of how they feed each other, and 45 engineering queries raised when the documents were checked against each other and against the simulator model.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
Prepared for	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
Document no.	PVC-V1-PM-DOC-001
Title	Document register and engineering queries
Status	Rev 0 · Issued for teaching
Originator	Apinan Soottitantawat

Rev	Date	Description	Originator	Checked	Approved
0	2026-10-27	Issued for teaching	Apinan Soottitantawat		

The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Document register

Document no.	Title	Discipline	Type	Rev	Status	File	Source of the numbers
PVC-V1-PM-DOC-001	Document register and engineering queries	Project	Register	0	Issued for teaching	PVC-V1-PM-DOC-001.pdf	This tab
PFD-100	Process flow diagram, R-101 area	Process	Drawing	0	Issued for teaching	pfd-100.pdf	pfd-a4-drawing from the model
PVC-V1-PE-HMB-001	Heat and material balance, line sizing, hydraulics	Process	Calculation	0	Issued for teaching	PVC-V1-PE-HMB-001.pdf	Model runs (reference batch, design state, 2 off-design cases)
PVC-PID-001	P&ID: reactor, charges, vents, SIS	Process / I&C	Drawing	A	Issued for teaching	PVC-PID-001.pdf	EPC sheet (alternative version on the P&ID tab)
PVC-PID-002	P&ID: heat removal	Process / I&C	Drawing	A	Issued for teaching	PVC-PID-002.pdf	EPC sheet (alternative version on the P&ID tab)
PVC-V1-ME-EQL-001	Equipment list and datasheets	Mechanical	List	0	Issued for teaching	PVC-V1-ME-EQL-001.pdf	Model, PFD-100, Design tab sizing
PVC-V1-PE-RLB-001	Relief load basis	Process safety	Calculation	0	Issued for teaching	PVC-V1-PE-RLB-001.pdf	Model relief runs (SIS off), API 520/521
PVC-V1-IN-INL-001	Instrument list and I/O count	I&C	List	0	Issued for teaching	PVC-V1-IN-INL-001.pdf	DCS bubbles, ALM, trips, fail positions
PVC-V1-IN-CE-001	Cause and effect diagram (A3)	I&C	Diagram	R0	Issued for teaching	PVC-V1-IN-CE-001.pdf	SRS/C&E data (sf.json)
PVC-V1-IN-SRS-001	Safety requirements specification of the SIFs	I&C / safety	Specification	R0	Issued for teaching	in the OTS	In the SRS of SIF tab (CSV download)
PVC-V1-SA-HZP-001	HAZOP report	Process safety	Study report	A	Issued for teaching	PVC-V1-SA-HZP-001.pdf	HAZOP tab data
PVC-V1-EE-LL-001	Electrical load list, cables, hazardous area	Electrical	List	0	Issued for teaching	PVC-V1-EE-LL-001.pdf	Model motor powers, IEC 60079-10-1 (typical)
PVC-V1-PM-CSU-001	Procurement, completion and commissioning plan	Project	Plan	0	Issued for teaching	PVC-V1-PM-CSU-001.pdf	Equipment and instrument lists, PROCS, SOP register
SOP-R101-01…12	Standard operating procedures (12)	Operations	Procedures	—	In the OTS	in the OTS	SOP tab
—	HAZID register	Process safety	Study	—	In the OTS	in the OTS	HAZID tab
—	LOPA and bow-ties	Process safety	Study	—	In the OTS	in the OTS	LOPA tab
—	Safety-layers register	Process safety	Study	—	In the OTS	in the OTS	Safety layers tab
PVC-V1-PL-LAY-001	Plot plan, elevations, isometrics and the walking route	Layout / piping	Drawing set	0	Issued for teaching	PVC-V1-PL-LAY-001.pdf	EQL footprints, typical spacing rules (assumed), HMB line list
—	Single-line diagram, loop diagrams, cable schedule	Electrical / I&C	Drawings	—	Not issued	in the OTS	

2 Document flow

From	To	What passes
PFD-100	PVC-V1-PE-HMB-001	streams, numbers, equipment
PVC-V1-PE-HMB-001	PVC-V1-ME-EQL-001	operating P/T, flows, duties
PVC-V1-PE-HMB-001	PVC-V1-PE-RLB-001	reaction heat, vapour pressure, flows

From	To	What passes
PVC-V1-ME-EQL-001	PVC-V1-PE-RLB-001	MAWP, volumes, wetted area
PVC-PID-001	PVC-V1-IN-INL-001	tags, loops, fail positions
PVC-V1-IN-INL-001	PVC-V1-IN-CE-001	trip transmitters, final elements
PVC-V1-SA-HZP-001	PVC-V1-IN-SRS-001	SIF needs (via LOPA)
PVC-V1-ME-EQL-001	PVC-V1-EE-LL-001	motor ratings
PVC-V1-IN-INL-001	PVC-V1-PM-CSU-001	loop checks, I/O
PVC-V1-ME-EQL-001	PVC-V1-PM-CSU-001	procurement packages

3 Engineering queries

Status of every query: Open. Raised by the discipline named; "also" lists a second discipline that raised the same point.

EQ	Subject	Finding	Raised against	Proposed resolution	Raised by
EQ-01	P-103 power	econ() charges 55 kW for the VCM charge pump. The model's own numbers (180 t/h delivered at K.Pvcm = 13 bar a from TK-100 stored saturated at 3.98 bar a, 25 °C) need about 8.7 bar differential, 98 m head and ≈ 69 kW absorbed at an assumed efficiency of 0.70.	Model econ() / PVC-V1-EE-LL-001 (load list) / PVC-V1-ME-EQL-001	Rate P-103 for ≈ 70 kW absorbed (75 kW motor) or confirm a lower delivery pressure; carry the same figure into the equipment and load lists.	Process
EQ-02	P-101A/B power	econ() charges 90 kW per running jacket pump; the loop Δp assumed here (jacket 2.2 bar + mixer + valves + 90 m of 10" pipe ≈ 2.9 bar, 30 m) gives ≈ 48 kW absorbed. Either the jacket Δp is about twice the assumption or 90 kW is a motor rating, not a load.	Model econ() / PVC-V1-ME-EQL-001 (jacket Δp) / PVC-V1-EE-LL-001	Confirm the jacket pressure drop at 470 m³/h on the vessel datasheet; state in the load list whether 90 kW is the rating or the absorbed load.	Process
EQ-03	Degassed VCM in the Process tab stream table	Process tab stream 8 'Degassed VCM' gives the VCM left in R-101 at the end point (≈ 7.8 t). Degassing to 1.5 bar g sends 6.25 t to recovery; ≈ 1.5 t stays dissolved in the PVC and the water and leaves with the slurry to D-102 and the stripper (Transition tab: 6.23 t vented, 1.48 t in the slurry).	Process tab (stream table, stream 8); Design tab text '≈ 15 % of the charge goes back to recovery'	Show stream 8 as 6.2 t/batch to recovery and add the VCM in the slurry (1.5 t) to stream 9; quote ≈ 12 % of the charge to recovery.	Process
EQ-04	Utility streams missing on PFD-100	The shrinkage-water injection (FV-103), the CW to TV-110B, the baffle CW (FV-120), the E-101 CW (FV-130) and the N_2 pad have no PFD-100 stream numbers (note 3 of the drawing omits baffle CW and the N_2 pad). The HMB carries them with the Process tab numbers only.	PFD-100 rev 0	Number the utility streams on PFD-100 rev 1 (or add a utility flow diagram) so the HMB and the line list can reference one number.	Process
EQ-05	Cooling margin on a warm day	Heat-removal capacity at the peak state with every handle wide open is 1.34 × the peak heat release at CW 32 °C but only 1.01 × at 38 °C; off-design A drives FV-130 to 540 m³/h (wide open) and TV-110B to 100 %. The cooling is sized at the 32 °C design CW temperature with no allowance for summer.	Design tab (design basis) / Operating windows (CW range)	Confirm the CW design supply temperature for the site (summer); if 38 °C occurs, either add E-101 area or reduce the initiator dose on warm days (procedure).	Process
EQ-06	Cycle-time basis	The plant capacity (200 kt/y) uses a 6.9 h cycle (model card). The scripted cycle of this HMB is 7.7 h for steps 1–8 (Transition tab cold batch 7.75 h) plus the rinse, i.e. about 170 kt/y with four reactors at 8 000 h/y.	Model card / RESUME basis; Transition and Process reliability tabs	Fix one cycle-time basis for all documents (already an open item between Process reliability 6.9 h and Transition 7.75 h).	Process
EQ-07	R-101 jacket design pressure	P&ID-001 note 3 gives the jacket 5.2 bar g, but the loop runs at ≈ 7.1 bar g at the P-101 discharge (CW return 2 bar g + 5.1 bar implied by the 90 kW pump power in the model) and ≈ 8.1 bar g at shut-off.	PVC-PID-001 note 3; PVC-V1-ME-EQL-001; model econ() P-101 power	Raise the jacket design to 10 bar g (typical for tempered-water jackets) and re-check the shell for external pressure (it then needs a thicker shell or stiffening), or confirm a P-101 head below 3.2 bar and lower the pump power in the model.	Mechanical (also erel)
EQ-08	R-101 shell thickness governed by external pressure	A 30 mm shell (internal pressure, UG-27) buckles under jacket 5.2 bar g + full vacuum (Pa 2.9 bar < 6.2 bar). The shell needs 40 mm with a 7.4 m unstiffened jacket.	PVC-V1-ME-EQL-001 R-101 datasheet	Vendor to choose between a thicker shell, jacket closure/stiffening rings or a half-pipe jacket; the wall resistance enters the jacket U (model U0 0.75 kW/m² K): confirm U with the final wall.	Mechanical
EQ-09	P-103 shut-off pressure above the R-101 design pressure	With TK-100 at 50 °C the VCM charge pump can reach ≈ 17.6 bar g at shut-off, above the 15.2 bar g design of R-101: a liquid-full reactor on the VCM charge is an overpressure case.	PVC-V1-PE-RLB-001 (relief basis); PVC-V1-ME-EQL-001	Add the liquid-overfill case (P-103 at shut-off) to the relief basis of PSV-100, or limit the pump curve / add a P-103 discharge PSV; LSHH-100 stays the first protection.	Mechanical
EQ-10	Equipment not in the model	Sizes, design conditions and motors of TK-107, D-104, P-104, D-105, D-103, D-106, C-101, P-102 and the motor ratings of M-101, P-101A/B, VP-101 are assumed in this document.	PVC-V1-ME-EQL-001	Confirm with the process data sheets (Process engineering tab) and the relief basis (D-103 volume, HAZOP R08).	Mechanical
EQ-11	PSV-100 capacity in the model vs the relief load	The OTS model relieves PSV-100 with nPsv = 0.25·√(P − 1.2) kmol/s, i.e. 227 t/h at 10 % overpressure (≈ 12599 mm², one T orifice). The governing case (Jacket temperature control fails to full steam (TV-110A wide open) at the design state, SIS off) needs 346 t/h of vapour at the set, and in that run the model PSV lets R-101 reach 18.71 bar g, above MAWP + 10 % (16.72 bar g).	Model (step()) and PSV-100 sizing	Size PSV-100 for 346 t/h: 19182 mm² → 2 × R (6 × 8 in, conventional, with RD-100); update the relief coefficient in step() to the selected capacity (372 t/h at 16.5 bar g) and re-run the relief cases.	Relief & vent

EQ	Subject	Finding	Raised against	Proposed resolution	Raised by
EQ-12	Two-phase relief of the runaway (DIERS) not done	A VCM suspension relieves as a foamy two-phase slurry; the model and this basis use vapour only, which gives the smallest possible area. HAZOP R08 is still open.	PVC-V1-PE-RLB-001, HAZOP R08	Two-phase (DIERS, e.g. homogeneous-equilibrium / omega method) sizing of PSV-100 / emergency vent and of the line to D-103 from calorimetry data; keep the vapour figures as the lower bound.	Relief & vent
EQ-13	D-103 outlet and capacity	PFD-100 shows PSV-100 and XV-108 discharging to D-103 but no outlet from D-103. The model runs relieve up to 26.3 t (CW loss · start) as vapour and the ESD vent sends ≈ 23 t (Transition tab); a two-phase relief could carry up to the whole liquid inventory (121.3 m^3 at the design state).	PFD-100, Mechanical (D-103), Relief basis	Show the D-103 vapour route (to VCM recovery / gasholder D-106 or vent treatment), size D-103 for one full two-phase relief plus quench water, and give D-103 its own design pressure ≥ the PSV-100 back-pressure.	Relief & vent
EQ-14	Temperature during a relieving runaway above the design temperature	With the SIS off the batch reaches 106.3 °C (CW loss, 4.85 bar g at that moment) and 112.6 °C (TV-110A full open, 1.48 bar g) — above the 93 °C design temperature of R-101.	Mechanical (R-101 design conditions)	Add a coincident design condition (pressure at the peak temperature) to the R-101 datasheet or confirm the vessel is acceptable at that temperature; this is an upset, not the design point.	Relief & vent
EQ-15	Short-stop pot D-105 pressure	D-105 must sit above the trip pressure (PSHH-101 12.5 bar g) plus line losses to inject; its pressure, its N_2 supply and any restriction orifice are not stated. If its N_2 supply exceeds the 15 bar g set, gas blow-through after the injection is a PSV-100 scenario.	P&ID-001, Process	State the D-105 operating / design pressure and the N_2 supply pressure; add a restriction orifice and include the blow-through case in the relief basis if the supply is above the set.	Relief & vent
EQ-16	Overfill: margin between the injection pump shut-off and the PSV set	With FV-103 wide open the vessel goes liquid-full and the pressure stops at the injection-pump shut-off (K.Pinj 16 bar a = 14.99 bar g): 14.98 bar g, only 0.02 bar below the set.	Process (injection pump), Relief basis	Confirm the injection-pump curve; if its shut-off can exceed 15 bar g, add the liquid case to PSV-100 (≈ 10 m^3/h, see 4.2) and keep LSHH-100 independent (HAZOP R16) with a second shut-off valve (R17).	Relief & vent
EQ-17	Power failure reaches the PSV set after the SRS run window	The SRS run of power failure stops at 60 min (peak 10.39 bar g). The same case run for 3 h with the SIS off lifts PSV-100 at 77:32 (TSHH point 55:02, PSHH point 72:48).	SRS of SIF (PVC-V1-IN-SRS-001), SIF-2/SIF-3 basis	Extend the SRS demand run to the PSV set (≥ 90 min) and state the process safety time for power failure; essential power for M-101 and P-101A/B (HAZOP R02) stays the main measure.	Relief & vent
EQ-18	LSHH-100 sensor	PVC-PID-001 draws LSHH-100 on LT-100, the LIC-100 control radar; the model trips on its own level measurement (LI101). The INL lists the drawing, so the SIS trip shares the control transmitter.	PVC-PID-001, PVC-V1-IN-SRS-001 (SIF-4)	Add an independent level transmitter LT-101 for LSHH-100 (SRS proposal LSHHB) and redraw PVC-PID-001.	Instrumentation & control
EQ-19	Pressure transmitter range	PT-100 and PT-101 are ranged 0–16 bar g but R-101 is evacuated to 0.03 bar a (−0.98 bar g) before every batch; the evacuation permissive reads PI-100.	IND / DCS ranges, PERM-EVAC (C&E)	Range PT-100 as a compound transmitter −1 to 16 bar g (or add an absolute-pressure transmitter for the evacuation); keep PT-101 0–16 bar g for the SIS if the evacuation check stays on PT-100.	Instrumentation & control
EQ-20	TIC-100 tuning	PVC-PID-002 note 3 gives Kc 10, Ti 1500 s, Td 120 s; the model runs Kc 12, Ti 900 s, Td 120 s (retuned after the drawing was issued).	PVC-PID-002 rev A	Update note 3 to the model tuning at the next revision, or move tuning to the DCS configuration database (tuning is not normally a P&ID note).	Instrumentation & control
EQ-21	Energise-to-trip short-stop and vent valves	XV-104 and XV-108 must open on a trip but are drawn fail-closed: loss of air or power defeats the action (already in C&E note 12 and HAZOP R39).	PVC-PID-001, PVC-V1-IN-SRS-001	Dedicated air receiver with low-pressure alarm and line monitoring of the solenoid circuits (energise-to-trip design per IEC 61511), or a fail-open design for XV-104 with a separate isolation.	Instrumentation & control
EQ-22	Air failure in the model	The model does not move XV-103, XV-104, XV-108 on instrument-air failure, while the drawings show them fail-closed.	Simulator model (step())	Model all drawn fail positions on air failure (OTS change request), then rerun the air-failure checks.	Instrumentation & control
EQ-23	SIS action A	drawn: close XV-101, XV-102, XV-103; model closes XV-101, XV-102, XV-109 (differ: XV-103, XV-109).	PVC-PID-001 SIS table	Correct the SIS table on PVC-PID-001 to the C&E (XV-109 closes, XV-103 is a permissive) at the next revision.	Instrumentation & control
EQ-24	HIC-135 / HIC-105 flows	The faceplates show the inert-vent flow (kmol/h) and the degassing flow (t/h), but PVC-PID-001 draws no flow element on either line; the values are estimates.	PVC-PID-001, DCS displays	Either add FT-135 / FT-105 (vortex or thermal mass) or label the values as estimated on the displays.	Instrumentation & control
EQ-25	PSV-100 lift alarm	The C&E lists a PSV-100 lift alarm, but no lift switch or rupture-disc indicator is drawn; the DCS can only infer the lift from PI-100.	PVC-V1-IN-CE-001, PVC-PID-001	Add a lift / burst indicator (e.g. ZSO-100) on PSV-100 or its rupture disc and wire it to the DCS.	Instrumentation & control

EQ	Subject	Finding	Raised against	Proposed resolution	Raised by
EQ-26	Non-ISA first letters	KX-100 uses K (time, schedule in ISA-5.1) for the K-value, VPC-130 uses V (vibration) for a valve-position controller, and QI-100 (quantity) carries a heat duty.	DCS tag list	Keep the tags (they are on every display and in the training material) but list the meaning in the INL legend; for a new unit use AY-100 (property, computed), ZC-130 and JY-100.	Instrumentation & control
EQ-27	Agitator power: model "full load" vs the batch peak	K.agitKW is 290 kW, but step() scales it by (0.9 + 0.25 X) and the level, so the reference batch draws 290 kW at the design state and 338 kW at the end point (X 0.848). With gearbox η 0.97 and 15 % margin (assumed) M-101 needs 401 kW → 450 kW IEC.	Model (K.agitKW label), Mechanical (M-101 motor rating)	State M-101 as 450 kW rated / 338 kW absorbed peak in the equipment list and datasheet; keep K.agitKW as the base power in the model notes, not "full load".	Electrical (also ecomm)
EQ-28	Essential power supply not shown	HAZOP R02 asks for M-101 and P-101A/B on essential power; the load list puts M-101, P-101A, D-104 CH, UPS-101 (running loads) on an essential bus of 535 kVA running load (M-101 DOL start on top). No standby generator or essential switchboard appears on PFD-100 or P&ID-001.	Electrical single-line (not yet issued), P&ID-001, HAZOP R02	Add an essential board fed by a standby generator sized for the essential load (≥ 669 kVA with 25 % margin, typical) and for the M-101 start; prove by test that the short-stop is mixed after a power failure.	Electrical
EQ-29	UPS autonomy vs the power-failure horizon	With the SIS failed, a power failure at the design state reaches the PSV-100 set after 77:32 (Relief & vent run 3.4); a typical 30-min UPS keeps the SIS and DCS alive for less than that if the essential supply also fails.	Electrical (UPS specification), SRS of SIF	Specify UPS autonomy ≥ 80 min for the SIS, DCS and the short-stop solenoid, or put the UPS charger on the essential (generator) bus and state it in the SRS.	Electrical
EQ-30	VCM auto-ignition temperature differs between sources	ICSC 0082, NOAA CAMEO, INRS FT 184 and DOE-HDBK-1092 give an auto-ignition temperature of 472 °C (T1); the IISc ENVIS data sheet gives an ignition temperature of 415 °C (T2). The temperature class must be chosen on the lower value.	PVC-V1-EE-LL-001 (hazardous-area classification)	Classify the R-101 area IIA T2; specify equipment T3 (200 °C) or better so the choice does not depend on the source.	Electrical
EQ-31	C-101 recovery compressor duty	C-101 is shared by R-101…R-104 and has no power in the model. The load list estimates 210 kW from the residual VCM at the end point (7.76 t) degassed in 1 h (assumed) from 1.25 to 6.05 bar a.	Process (degassing rate, compressor sizing), Mechanical	Confirm the degassing schedule of the four reactors (staggered or not) and the C-101 design flow; resize the motor and its feeder.	Electrical
EQ-32	Hazardous-area extents are typical only	Zone extents use typical distances: the ventilation of the reactor bay, the layout and the release rates of the sources (IEC 60079-10-1) are not yet defined.	Layout (step 1.6), Safety	Calculate release rates and the ventilation degree per IEC 60079-10-1 once the layout of the reactor bay is fixed; issue the hazardous-area drawing.	Electrical
EQ-33	Cooling water to the jacket not metered	The CW injected into the jacket loop through TV-110B (and spilled through PCV-112) has no flowmeter; the performance test can only estimate it from QI-100 and the baffle and condenser meters.	PVC-PID-002, PVC-V1-IN-INL-001	Add a flow element on the CW injection or the spill line if the CW consumption becomes a guarantee; otherwise record it as an estimate.	Commissioning
EQ-34	Procedure key “esd”	The OTS procedure with key “esd” is the jacket-pump switch-over (keep the batch), not an emergency shutdown; the commissioning plan maps it to the pump runs.	OTS procedure list (PROCS)	Rename the key (e.g. “pumpfail”) at the next OTS revision; keep the title.	Commissioning
EQ-35	Oxygen acceptance by pressure only	R-101 has no oxygen analyser: the acceptance before VCM is the evacuation pressure (permissive 0.32 bar a). Commissioning has no independent check of the residual O_2 after the first evacuation.	PVC-V1-IN-INL-001, SOP-R101-02	Use a portable O_2 analyser on the vacuum line for the first batches (commissioning only), or add an O_2 analyser on the vacuum-pump discharge.	Commissioning
EQ-36	Unnumbered spools	Four spools drawn on the isometrics have no number on the line list EP_HMB.lines: U-01 jacket return J2 → P-101A/B suction header (10"), U-13 jacket-loop low-point drain with HV-261, U-09 R-101 bottom drain with HV-260, U-07 PSV-100 inlet spool with HV-250 / RD-100. layout.json also carries U-02, U-03 (CW returns), U-08 (XV-108 vent), U-10 (initiator), U-11 (short-stop, HV-252), U-12 (PVA, HV-107), U-14 (vacuum), U-16 (D-106 → C-101).	PVC-V1-PE-HMB-001 line list	Give every spool that carries a hand valve or a relief device a line number (jacket return, PSV inlet, drains, short-stop, vacuum, CW returns) and size it in the line list.	Layout
EQ-37	P-103 hydraulics vs layout	The P-103 calculation assumes a static rise of 10 m, a discharge line of 80 m and a suction line of 30 m (EP_P103). The layout gives 18.55 m (P-103 discharge EL +1.2 to the R-101 top nozzles EL +19.75), 120 m of L-103 and 11.8 m of L-102. NPSHa rises from 3.83 to 3.89 m; the pressure drop left for FV-102 at the highest reactor pressure falls from 3.41 to 2.49 bar (still ≥ 0.5 bar).	PVC-V1-PE-HMB-001 section 5 (pump hydraulics)	Re-run the P-103 hydraulics with the layout lengths and elevations; check the FV-102 sizing at the end of the charge.	Layout

EQ	Subject	Finding	Raised against	Proposed resolution	Raised by
EQ-38	L-115 size	The line list gives no size for L-115 (PSV-100 discharge); layout.json and ISO-6 assume 8" (same as the R-101 N7 nozzle).	PVC-V1-PE-HMB-001 / PVC-V1-PE-RLB-001	Size L-115 and the inlet spool from the relief load (built-up back pressure and inlet loss), then update the isometric.	Layout
EQ-39	PSV-100 inlet spool length	With PSV-100 on the top floor F4 (EL +23.0) the inlet spool from the top-head nozzle (EL +19.75) through HV-250 and RD-100 is 4.25 m long.	PVC-V1-PE-RLB-001	Check the inlet pressure loss against the typical 3 % of set pressure (API 520 Part II practice); if it fails, move PSV-100 down to F3 or enlarge the spool.	Layout
EQ-40	R-101 support and elevations	The Mechanical GA sketch shows R-101 on a skirt with the bottom at EL +3.0 (assumed). The layout needs the bottom valve 2.6 m above the D-102 inlet for gravity discharge and weighs the reactor with load cells, so R-101 hangs on four lugs on the F2 beams (EL +13.0) with the bottom-head apex at EL +9.875 and the thermowells at EL +14.0.	PVC-V1-ME-EQL-001 GA and nozzle list	Mechanical to adopt the layout elevations and support (lugs, no skirt) and update the GA and the vessel weight basis.	Layout
EQ-41	R-101 nozzle orientation	The GA nozzle plan (EM_PLAN) is schematic and has one nozzle N6 for vapour to E-101 and reflux return, while the line list has two lines (L-109 12", L-110 8"). The layout fixes the orientation with two separate nozzles (vapour at 30°, reflux at 0°) and the baffle CW as one inlet and one outlet distributor (N19/N20 are four each).	PVC-V1-ME-EQL-001 nozzle list	Issue a nozzle orientation drawing from the layout; split N6 into vapour and reflux nozzles or show the reflux entering through the vapour nozzle.	Layout
EQ-42	Local gauges	24 local gauges (PG/TG/LG/SG tags, e.g. PG-101A/B, TG-110, PG-250 tell-tale between RD-100 and PSV-100, LG-150) are placed for the field round but are not on the instrument list or the P&ID.	PVC-V1-IN-INL-001, PVC-PID-001/002	Add the local indicators to the P&ID and the instrument list (no I/O).	Layout
EQ-43	Field valves of the model	HV-107 (PVA pot) and HV-141 (N_2 pad) are field valves of the model (P.hv, set in the Field tab) but are not in HANDVALVES.json; layout.json carries them in an extension array 'fieldvalves'.	Hand-valve list of the model (Visual plant 3D tab) and PVC-PID-001/002	Decide whether HV-107/HV-141 join the hand-valve list (P.hm) or stay as P.hv valves; Visual plant 3D reads 'fieldvalves'.	Layout
EQ-44	VCM charge permissive does not check the degassing valve	The batch logic refuses the VCM charge (XV-102) while the vacuum valve XV-109 is open, but not while the degassing valve XV-105 is open; Tavaux 2004 (Incident cases tab) released 1.6 t of VCM that way.	Batch logic (bmsCmd 'vcm'), PVC-V1-IN-CE-001 permissive rows, HAZOP	Add "XV-105 closed" to the VCM-charge permissive and to the C&E permissive rows; review in the HAZOP procedures node.	Project
EQ-45	PSV-100 lift time quoted twice	For the passing injection-water case the LOPA tab notes give PSV lift at 165:30 while the SRS tab and its replay give 164:50–164:55 (different run settings).	LOPA tab, PVC-V1-IN-SRS-001	Quote one run with its settings in both documents.	Project

4 Cross-document checks

Check	Value	Criterion	Result
Process engineering: document data on this page	EP_HMB present	the tab and its PDF read one data source	✓
Mechanical: document data on this page	EM_EQL present	the tab and its PDF read one data source	✓
Relief & vent: document data on this page	ER_RLB present	the tab and its PDF read one data source	✓
Instrumentation & control: document data on this page	EI_INL present	the tab and its PDF read one data source	✓
Electrical: document data on this page	EE_LL present	the tab and its PDF read one data source	✓
Procurement to commissioning: document data on this page	EC_PLAN present	the tab and its PDF read one data source	✓
Document numbers unique	15 numbers	no number used twice	✓
Every registered document opens a tab that exists	all present	tab id found on the page	✓
PSV-100 set ≤ R-101 design pressure (RLB vs EQL)	15 vs 15.2 bar g	set ≤ MAWP	✓
PSV-100 set = model K.Ppsv	15 vs 15 bar g	same number	✓
Motor ratings: equipment list = load list	9 motors agree	same rated kW	✓

Check	Value	Criterion	Result
Instrument list trip settings = model K	4 trips agree	TSHH 63 °C, PSHH 12.5 bar g, FSLL 120 m³/h, LSHH 94 %	✓
HMB charges = model recipe	water 82.0 t, VCM 51.0 t	water ≥ 60 t (+ injection), VCM 51 t	✓
Every OTS procedure has a commissioning step	6 of 6	each PROCS key used in the plan	✓
Every engineering query names the document it is raised against	45 queries	a document or study in the register	✓
Open engineering queries	45	all closed before "issued for construction" (here: issued for teaching, so open is expected)	info

Document control: revision letters before IFC and numbers after; status IFR → IFA → IFC → as-built. Every document of this OTS is Rev 0 (P&ID rev A) · Issued for teaching and must not be used for construction.